



EMPOWERING FUTURE



MECWIN – EMPOWERING THE FUTURE OF AEROSPACE & DEFENCE

About Company

Mecwin is a leading provider of sophisticated energy and motion solutions, including solar pumps, MPPT controllers, PMSM motors, fluid pumps, and electric vehicle motors, tailored for critical applications in the medical, defence, and aerospace sectors. With a strong emphasis on innovation, design, development, and manufacturing, we deliver high-performance, reliable technologies that optimize energy systems and industrial operations while advancing sustainable development. Serving the renewable energy, automation, agriculture, and industrial sectors, Mecwin consistently sets benchmarks in efficiency, reliability, and technological excellence, empowering industries worldwide to achieve exceptional performance and long-term growth.



Our Vision

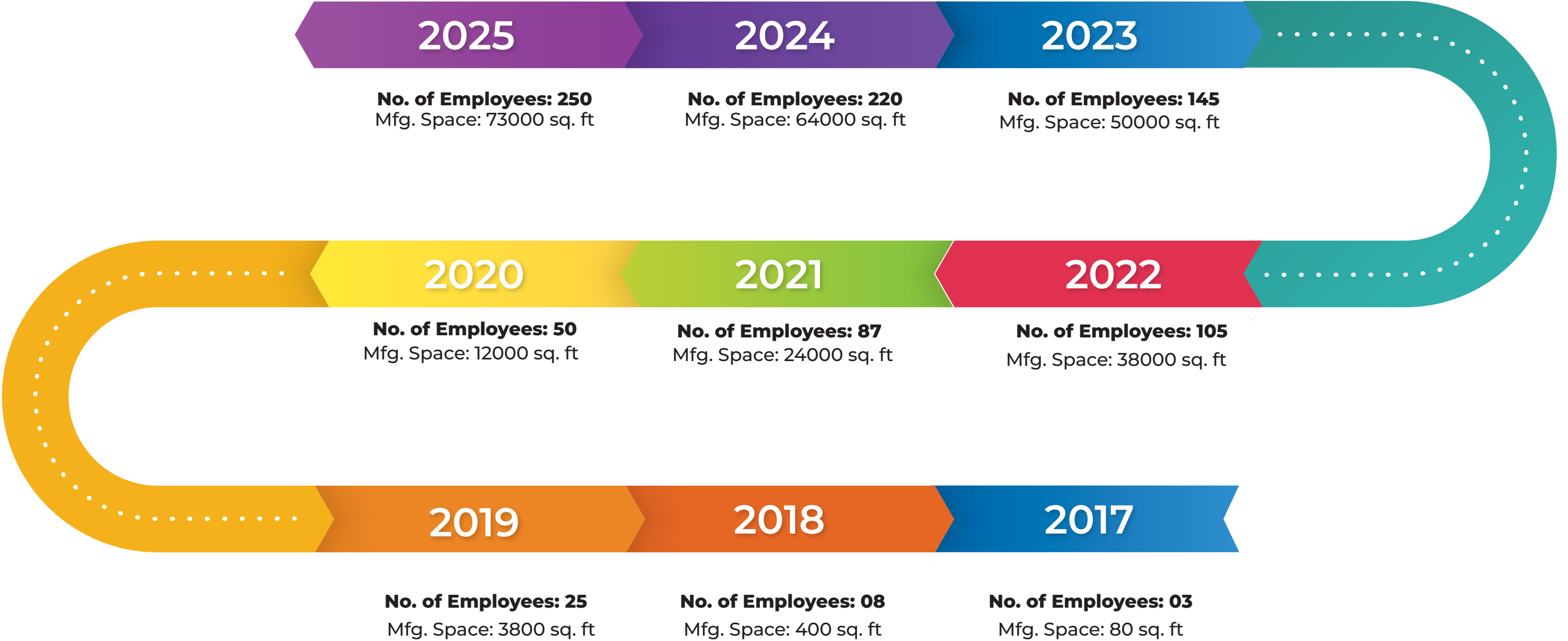
At Mecwin Technologies, we stand at the threshold of a profound mission—one that transcends mere corporate ambitions and resonates with the very essence of our existence. “Innovating for a Greener World: Unleashing Sustainable Possibilities” is not just a statement; it is a resounding call to action, a promise to the planet, and an unwavering commitment to the future.

Our Mission

At Mecwin Technologies, our mission is to lead the way in transforming our world into a greener, more sustainable place. We are dedicated to innovating and delivering solutions that not only meet the needs of today but also preserve the planet for future generations. In addition, we design, develop, and manufacture motors and actuators for aerospace and defence applications, focusing on indigenization to support self-reliance and cutting-edge technology.



Milestone



Why Choose Mecwin ?



End-to-End Capability

Comprehensive in-house design and manufacturing of BLDC, PMSM, and special-purpose motors, paired with advanced controllers and smart IoT-enabled platforms



Diverse Product Portfolio

Mid-Drive Motors, Hub Motors, Traction Motors, Industrial Pump Motors, Customized Induction Motors, PMDC Motors, BLDC Motors, PMSM Motors & Motor Controller Solutions



Make in India Advantage

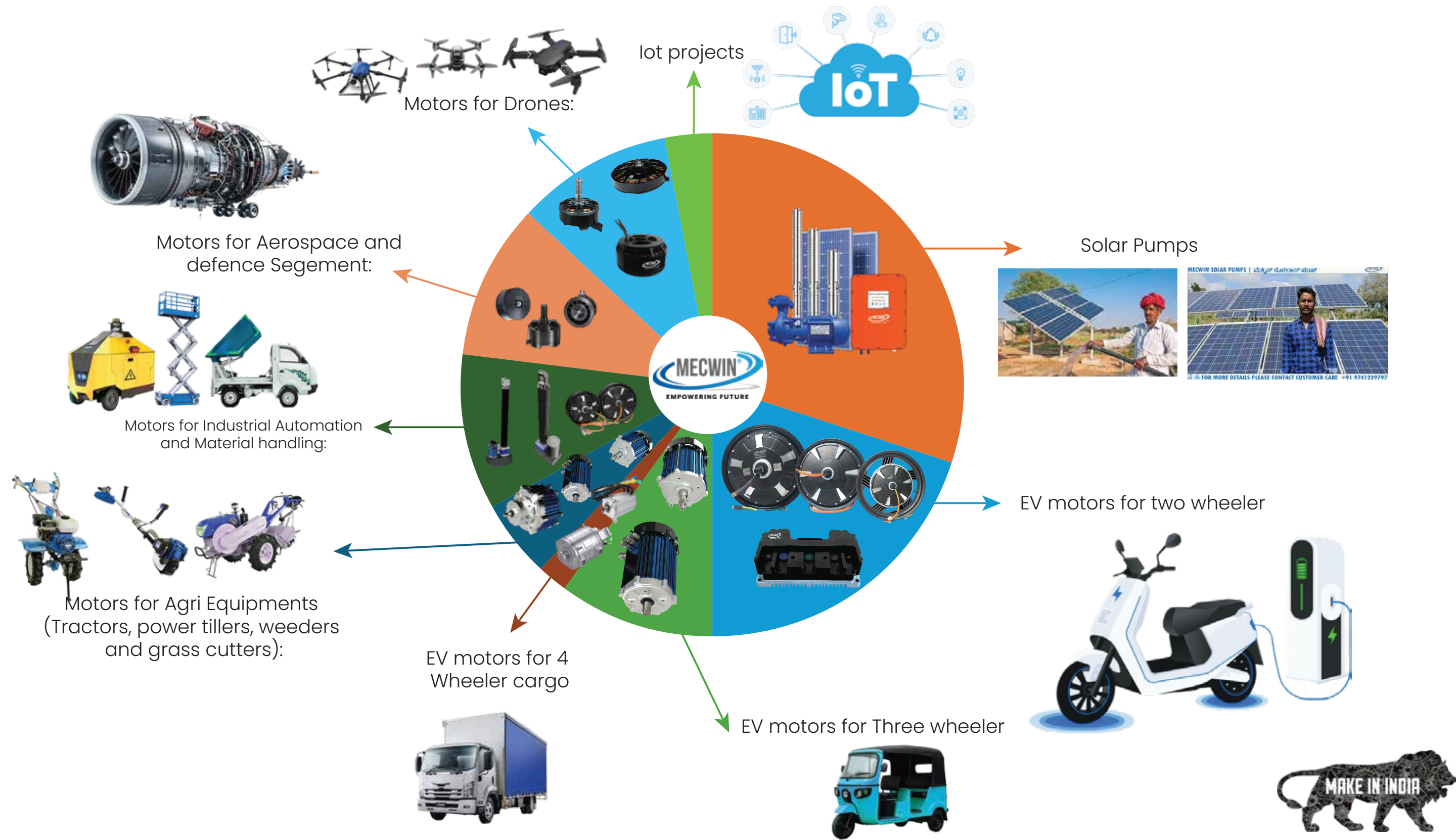
Fully localized production with deep expertise in motor design, power electronics, and system integration, reducing reliance on imports



Strong Talent Pool

Experienced team of engineers and technologists with domain knowledge in automotive, powertrain, defence, medical, industrial and renewable energy systems.





Product Developed for Various Aerospace & Defence Programmes :

VTOL Motor



VTOL motors are high-performance electric or hybrid propulsion systems that enable aircraft to take off and land vertically

Actuator Motor for Parachute Ejection while Landing



Actuator developed for automatic and reliable parachute deployment, minimizing impact forces and protecting both equipment and payload during landing.

Induction Motor for Scavenge Fan



Motor designed for the scavenge fan to remove residual hot air or gases from confined engine spaces, working with the main ventilation system to improve airflow and prevent heat buildup.

PMSM Motors for FPV Drones



The 2807 BLDC/PMSM drone motor is engineered for a wide voltage range, delivering fast response, high efficiency, and stable, precise flight performance

Induction Motors for Cooling Fan



Overall operational temperature limits: -55°C to + 70°C, Motor endurance >2000 hrs., MTBF >3000 hrs, MTBO >1000 hrs



Mecwin: Strengths in End-to-End Product Development & Qualification

Domain	Likely Motor/Actuator	Application	Notes / Assumptions
UAV	BLDC / PMSM	Propulsion motors, servo actuators	Long endurance drone; uses high power density motors for control surfaces and propulsion
Loitering Munition	BLDC	Main propulsion	Indigenous kamikaze drone; lightweight brushless motors used for low noise
Fighter Aircraft	Servo / BLDC / Starter-Generator	Flight control actuators, avionics cooling, APU	Uses multiple actuators for flaps, doors; APU starter-generator used
Helicopter	Induction / Starter Motor	Starter-generator, hydraulic motor drives	Known to use conventional motors in ground/start systems
Main Battle Tank	Induction / BLDC	Gun drives, turret motors, cooling fans	Uses motors for turret rotation, stabilisation, internal cooling
Missile	Servo Motor / Actuator	Fin control actuators	High-response torque motors needed; likely imported or custom built
Naval Ship	Induction / PMSM	Deck handling, radar systems, weapons lifts	Large vessels use industrial-class actuators and drives
Satellite	Stepper / BLDC / PMSM	Attitude control, solar panel deployment	Space-grade motors used in very small volumes; high reliability
Missile	Servo / BLDC	Control surface actuators	Autonomous navigation requires multiple compact actuators
UAV	BLDC / PMSM	Propulsion, flight actuation	Technology demonstrator for stealth UAV; uses low-RCS motors



Advanced Manufacturing Facilities for Drone Motor Production

Automatic Coil Winding Machine



Epoxy Potting Machine



CNC VMC Machine



CNC Turning Machine



Laser Engraving & Marking Machine



Comprehensive Qualifications Testing Facilities & Standard Compliances



Salt Spray Machine



Dust Ingress Machine



Environmental & Thermal Chamber



Vibration Shaker Interface Climatic Chamber



Motor Testing Dyno



Dynamic Balancing & Vibration Analyzer



Life Cycle / Endurance Test Rig

MIL-STD-810 [Environmental & Durability: Altitude, Thermal cycles, Humidity, Sand and Dust, Vibration.

MIL-STD-461 [MEI & EMC: Conducted Emissions (CE), Radiated Emissions (RE), Radiated Susceptibility (RS), Conducted Susceptibility (CS).

RTCA DO-160: The key DO-160 sections that apply to motors and actuators: Temperature and Altitude, Humidity, Vibration and Shock, Sand and Dust.

MIL-E-50070C: Vent Motors Environmental Tests: Environmental Conditions such as elevated ambient air temperatures, reflective radiant heat, dust, sand, and other environmental stressors.



Comprehensive Qualifications Testing Facilities & Standard Compliances

Motor Testing Dyno For R&D

Testing motor type: Hub motor & Mid motor & Axial flux motor

Technical Specifications:

- Maximum Power Capacity: 24 kW
- Maximum Speed: 10,000 RPM
- Maximum Torque: 300 N·m

Available Test Setups:

- Setup 1 – High Speed & Low Torque: 10,000 RPM & 10 N·m
- Setup 2 – Low Speed & High Torque: 1,000 RPM & 100 N·m

Basic Functional Tests Conducted :

- Acceleration and Deceleration Performance
- Grade Effect Analysis on Vehicle Performance
- Maximum Speed Validation
- Steady-State Speed Testing
- Vehicle Coast-Down Characteristics Test
- Range Testing
- Electrical Energy Consumption Measurement (using a Power Analyzer)
- Net Power Test
- Gradient Test (up to 30° inclination)
- Vehicle Power at Wheel Test (30-minute power test)
- Road Load Simulation Mode to replicate real-world load conditions inside the laboratory



Thrust Test Stand to Measure Motor & Propeller Performance



Direct measurements

- Torque (Nm)
- Thrust (kgf)
- Voltage (V) and current (A)
- Rotations per minute (RPM)
- Motor winding resistance (Ohm)
- Accelerometer on PCB (g)

Derived measurements

- Motor efficiency (%)
- Propeller efficiency (g/W)

Features

- USB interface
- ESC manual control
- Three servo control ports
- Automatic control
- Powerful scripting
- Three accessory ports
- Three temperature probe ports
- Output data to CSV file
- Real-time sensor plots
- Included calibration hardware

Applications

- Inrunner and outrunner brushless motor characterization (0–55 A)
- Propeller characterization
- Calculate system efficiency
- Servo testing and control
- Battery endurance testing
- Factory tests

Design specifications of the Series 1585 Test Stand

Specification	Min.	Max.	Tolerance	Unit
Thrust	-5	5	0.5% ±0.001	kgf
Torque	-2	2	0.5% ±0.001	Nm
Voltage	0	50	0.5%	V
Current	0	55	1%	A
Burst Current	0	60		A
Angular speed*	0	190k	1	eRPM
Coil resistance	0.003	240	0.5%	Ohm
Digital scale	0	3	0.5%	kgf

Electrical RPM, divide by the number of motor poles
to obtain true mechanical RPM

Note : At Mecwin, motors are tested on a thrust stand to evaluate their efficiency. The tests measure parameters including torque (Nm), thrust (kgf), voltage (V), current (A), rotations per minute (RPM), and motor winding resistance (Ω)...



Advanced Software Ecosystem for Product Innovation

Electromagnetic Design & Simulation



Mechanical Design & Analysis



ERP & Production Management Software



Mecwin Nethra RMS



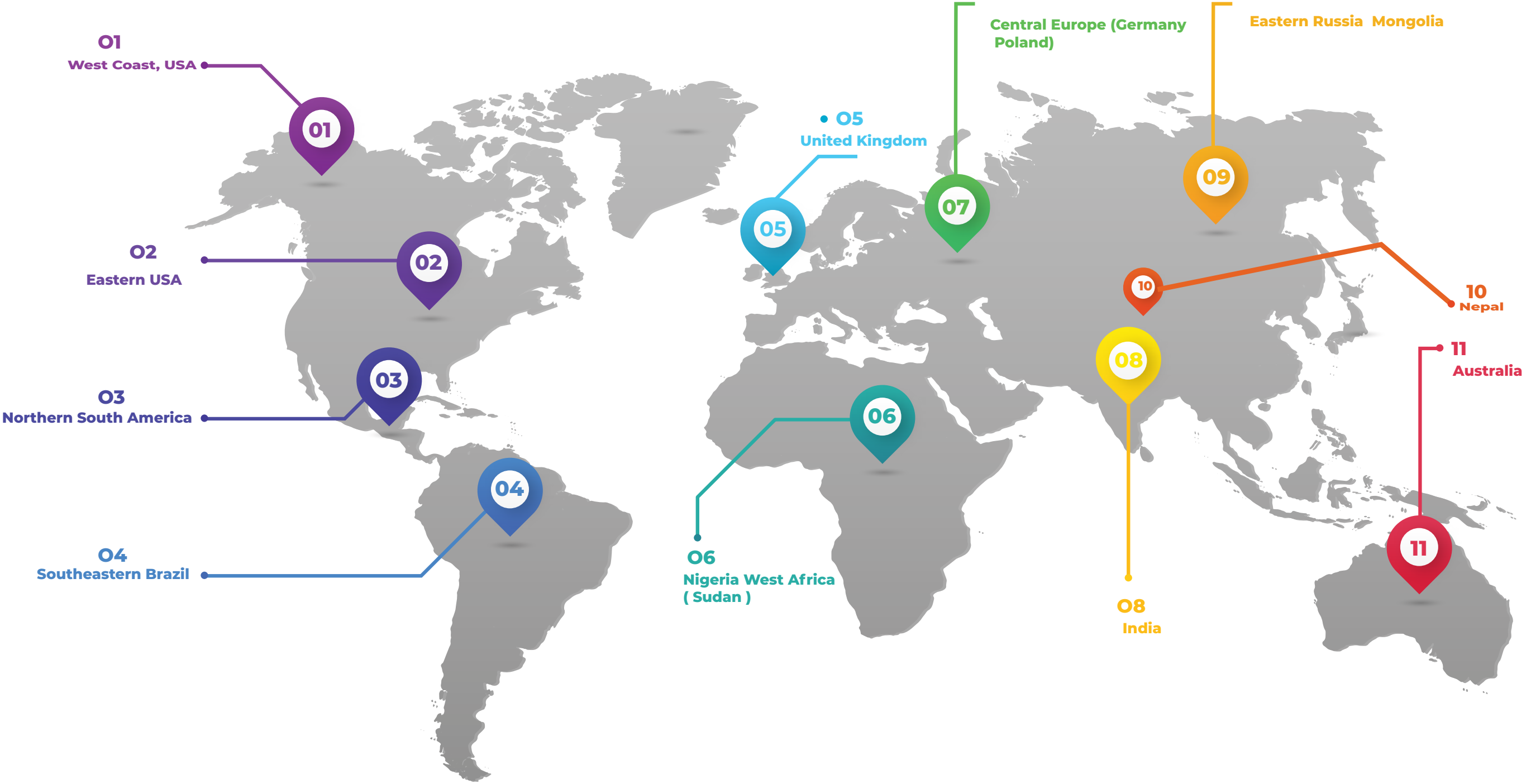
A Network of Service & Support



Our Service & Support team
are directly accessible in Pan India



MECWIN Global Customers



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